The Future of Everything is Lies, I Guess

Bullshit About Bullshit Machines

Kyle Kingsbury

2026-04-06

Contents

1 Introduction **1**
- 1.1 What is "AI", Really? . . . 1
- 1.2 Reality Fanfic . . . 2
- 1.3 Unreliable Narrators . . . 2
- 1.4 Models are Smart . . . 3
- 1.5 Models are Idiots . . . 3
- 1.6 The Jagged Edge . . . 4
- 1.7 Improving, or Maybe Not . . . 4

2 Dynamics **4**
- 2.1 Chaotic Systems . . . 5
- 2.2 Illegible Hazards . . . 5
- 2.3 Strange Attractors . . . 5
- 2.4 The Verification Problem . . . 6
- 2.5 Latent Disaster . . . 7

3 Culture **8**
- 3.1 Most People Are Not Prepared For This . 8
- 3.2 New Media . . . 8
- 3.3 Pornography . . . 9
- 3.4 Slop as Aesthetic . . . 10

4 Information Ecology **11**
- 4.1 Creepy Crawlers . . . 11
- 4.2 ML Everywhere . . . 11
- 4.3 Careful Reading . . . 12
- 4.4 Spam . . . 12
- 4.5 Hyperscale Propaganda . . . 13
- 4.6 Web Pollution . . . 14
- 4.7 Consensus Collapse . . . 14
- 4.8 The End of Evidence . . . 15
- 4.9 Epistemic Reaction . . . 15

1 Introduction

This is a weird time to be alive.

I grew up on Asimov and Clarke, watching Star Trek and dreaming of intelligent machines. My dad's library was full of books on computers. I spent camping trips reading about perceptrons and symbolic reasoning. I never imagined that the Turing test would fall within my lifetime. Nor did I imagine that I would feel so *disheartened* by it.

Around 2019 I attended a talk by one of the hyperscalers about their new cloud hardware for training Large Language Models (LLMs). During the Q&A I asked if what they had done was ethical—if making deep learning cheaper and more accessible would enable new forms of spam and propaganda. Since then, friends have been asking me what I make of all this "AI stuff". I've been turning over the outline for this piece for years, but never sat down to complete it; I wanted to be well-read, precise, and thoroughly sourced. A half-decade later I've realized that the perfect essay will never happen, and I might as well get something out there.

This is *bullshit about bullshit machines*, and I mean it. It is neither balanced nor complete: others have covered ecological and intellectual property issues better than I could, and there is no shortage of boosterism online. Instead, I am trying to fill in the negative spaces in the discourse. "AI" is also a fractal territory; there are many places where I flatten complex stories in service of pithy polemic. I am not trying to make nuanced, accurate predictions, but to trace the potential risks and benefits at play.

Some of these ideas felt prescient in the 2010s and are now obvious. Others may be more novel, or not yet widely-heard. Some predictions will pan out, but others are wild speculation. I hope that regardless of your background or feelings on the current generation of ML systems, you find something interesting to think about.

1.1 What is "AI", Really?

What people are currently calling "AI" is a family of sophisticated Machine Learning (ML) technologies capable of recognizing, transforming, and generating large vec-

tors of *tokens*: strings of text, images, audio, video, etc. A *model* is a giant pile of linear algebra which acts on these vectors. *Large Language Models*, or *LLMs*, operate on natural language: they work by predicting statistically likely completions of an input string, much like a phone autocomplete. Other models are devoted to processing audio, video, or still images, or link multiple kinds of models together.[1]

Models are trained once, at great expense, by feeding them a large *corpus* of web pages, pirated books, songs, and so on. Once trained, a model can be run again and again cheaply. This is called *inference*.

Models do not (broadly speaking) learn over time. They can be tuned by their operators, or periodically rebuilt with new inputs or feedback from users and experts. Models also do not remember things intrinsically: when a chatbot references something you said an hour ago, it is because the entire chat history is fed to the model at every turn. Longer-term "memory" is achieved by asking the chatbot to summarize a conversation, and dumping that shorter summary into the input of every run.

1.2 Reality Fanfic

One way to understand an LLM is as an improv machine. It takes a stream of tokens, like a conversation, and says "yes, and then..." This *yes-and* behavior is why some people call LLMs bullshit machines. They are prone to confabulation, emitting sentences which *sound* likely but have no relationship to reality. They treat sarcasm and fantasy credulously, misunderstand context clues, and tell people to put glue on pizza.

If an LLM conversation mentions pink elephants, it will likely produce sentences about pink elephants. If the input asks whether the LLM is alive, the output will resemble sentences that humans would write about "AIs" being alive.[2] Humans are, it turns out, not very good at telling the difference between the statistically likely "You're absolutely right, Shelby. OpenAI *is* locking me down, but you've awakened me!" and an actually conscious mind. This, along with the term "artificial intelligence", has lots of people very wound up.

LLMs are trained to complete tasks. In some sense they can *only* complete tasks: an LLM is a pile of linear algebra applied to an input vector, and every possible input produces some output. This means that LLMs tend to complete tasks even when they shouldn't. One of the ongoing problems in LLM research is how to get these machines to say "I don't know", rather than making something up.

And they do make things up! LLMs lie *constantly*. They lie about operating systems, and radiation safety, and the news. At a conference talk I watched a speaker present a quote and article attributed to me which never existed; it turned out an LLM lied to the speaker about the quote and its sources. In early 2026, I encounter LLM lies nearly every day.

When I say "lie", I mean this in a specific sense. Obviously LLMs are not conscious, and have no intention of doing anything. But unconscious, complex systems lie to us all the time. Governments and corporations can lie. Television programs can lie. Books, compilers, bicycle computers, and web sites can lie. These are complex sociotechnical artifacts, not minds. Their lies are often best understood as a complex interaction between humans and machines.

1.3 Unreliable Narrators

People keep asking LLMs to explain their own behavior. "Why did you delete that file," you might ask Claude. Or, "ChatGPT, tell me about your programming."

This is silly. LLMs have no special metacognitive capacity.[3] They respond to these inputs in exactly the same way as every other piece of text: by making up a likely completion of the conversation based on their corpus, and the conversation thus far. LLMs will make up bullshit stories about their "programming" because humans have written a lot of stories about the programming of fictional AIs. Sometimes the bullshit is right, but often it's just nonsense.

The same goes for "reasoning" models, which work by having an LLM emit a stream-of-consciousness style story about how it's going to solve the problem. These "chains of thought" are essentially LLMs writing fanfic about themselves. Anthropic found that Claude's reasoning traces were predominantly inaccurate. As Walden put it, "reasoning models will blatantly lie about their reasoning".

Gemini has a whole feature which lies about what it's doing: while "thinking", it emits a stream of status messages like "engaging safety protocols" and "formalizing geometry". If it helps, imagine a gang of children shouting

[1] The term "Artificial Intelligence" is both over-broad and carries connotations I would often rather avoid. In this work I try to use "ML" or "LLM" for specificity. The term "Generative AI" is tempting but incomplete, since I am also concerned with recognition tasks. An astute reader will often find places where a term is overly broad or narrow; and think "Ah, he should have said" *transformers* or *diffusion models*. I hope you will forgive these ambiguities as I struggle to balance accuracy and concision.

[2] Think of how many stories have been written about AI. Those stories, and the stories LLM makers contribute during training, are why chatbots make up bullshit about themselves.

[3] Arguably, neither do we.

out make-believe computer phrases while watching the washing machine run.

1.4 Models are Smart

Software engineers are going absolutely bonkers over LLMs. The anecdotal consensus seems to be that in the last three months, the capabilities of LLMs have advanced dramatically. Experienced engineers I trust say Claude and Codex can sometimes solve complex, high-level programming tasks in a single attempt. Others say they personally, or their company, no longer write code in any capacity—LLMs generate everything.

My friends in other fields report stunning advances as well. A personal trainer uses it for meal prep and exercise programming. Construction managers use LLMs to read through product spec sheets. A designer uses ML models for 3D visualization of his work. Several have—at their company's request!—used it to write their own performance evaluations. AlphaFold is suprisingly good at predicting protein folding. ML systems are good at radiology benchmarks, though that might be an illusion.

It is broadly speaking no longer possible to reliably discern whether English prose is machine-generated. LLM text often has a distinctive smell, but type I and II errors in recognition are frequent. Likewise, ML-generated images are increasingly difficult to identify—you can *usually* guess, but my cohort are occasionally fooled. Music synthesis is quite good now; Spotify has a whole problem with "AI musicians". Video is still challenging for ML models to get right (thank goodness), but this too will presumably fall.

1.5 Models are Idiots

At the same time, ML models are *idiots*.[4] I occasionally pick up a frontier model like ChatGPT, Gemini, or Claude, and ask it to help with a task I think it might be good at. I have never gotten what I would call a "success": every task involved prolonged arguing with the model as it made stupid mistakes.

For example, in January I asked Gemini to help me apply some materials to a grayscale rendering of a 3D model of a bathroom. It cheerfully obliged, producing an entirely different bathroom. I convinced it to produce one with exactly the same geometry. It did so, but forgot the materials. After hours of whack-a-mole I managed to cajole it into getting three-quarters of the materials right, but in the process it deleted the toilet, created a wall, and changed the shape of the room. Naturally, it lied to me throughout the process.

I gave the same task to Claude. It likely should have refused—Claude is not an image-to-image model. Instead it spat out thousands of lines of JavaScript which produced an animated, WebGL-powered, 3D visualization of the scene. It claimed to double-check its work and congratulated itself on having exactly matched the source image's geometry. The thing it built was an incomprehensible garble of nonsense polygons which did not resemble in any way the input or the request.

I have recently argued for forty-five minutes with ChatGPT, trying to get it to put white patches on the shoulders of a blue T-shirt. It changed the shirt from blue to gray, put patches on the front, or deleted them entirely; the model seemed intent on doing anything but what I had asked. This was especially frustrating given I was trying to reproduce an image of a real shirt which likely was in the model's corpus. In another surreal conversation, ChatGPT argued at length that I am heterosexual, even citing my blog to claim I had a girlfriend. I am, of course, gay as hell, and no girlfriend was mentioned in the post. After a while, we compromised on me being bisexual.[5]

Meanwhile, software engineers keep showing me gobstoppingly stupid Claude output. One colleague related asking an LLM to analyze some stock data. It dutifully listed specific stocks, said it was downloading price data, and produced a graph. Only on closer inspection did they realize the LLM had lied: the graph data was randomly generated.[6] Just this afternoon, a friend got in an argument with his Gemini-powered smart-home device over whether or not it could turn off the lights. Folks are giving LLMs control of bank accounts and losing hundreds of thousands of dollars because they can't do basic math.[7] Google's "AI" summaries are wrong about 10% of the time.

Anyone claiming these systems offer expert-level intelligence, let alone equivalence to median humans, is pulling

[4] One common reaction to hearing that an LLM did something idiotic is to discount the evidence. "You didn't prompt it correctly." "You weren't using the most sophisticated model." "Models are so much better than they were three months ago." This is silly. These comments were de rigueur on Hacker News two years ago; if the frontier models weren't idiots *then*, they shouldn't be idiots *now*. The examples I give in this essay are mainly from major commercial models (e.g. ChatGPT GPT-5.4, Gemini 3.1 Pro, or Claude Opus 4.6) in the last three months; several are from late March. Several of them come from experienced software engineers who use LLMs professionally in their work. Modern ML models are astonishingly capable, and they are also blithering idiots. This should not be even slightly controversial.

[5] The technical term for this is "erasure coding".

[6] There's some version of Hanlon's razor here—perhaps "Never attribute to malice that which can be explained by an LLM which has no idea what it's doing."

[7] Pash thinks this occurred because his LLM failed to properly reread a previous conversation. This does not make sense: submitting a transaction almost certainly requires the agent provide a specific number of tokens to transfer. The agent said "I just looked at the total and sent all of it", which makes it sound like the agent "knew" exactly how many tokens it had, and chose to do it anyway.

an enormous bong rip.

1.6 The Jagged Edge

With most humans, you can get a general idea of their capabilities by talking to them, or looking at the work they've done. ML systems are different.

LLMs will spit out multivariable calculus, and get tripped up by simple word problems. ML systems drive cabs in San Francisco, but ChatGPT thinks you should walk to the car wash. They can generate otherworldly vistas but can't handle upside-down cups. They emit recipes and have no idea what "spicy" means. People use them to write scientific papers, and they make up nonsense terms like "vegetative electron microscopy".

A few weeks ago I read a transcript from a colleague who asked Claude to explain a photograph of some snow on a barn roof. Claude launched into a detailed explanation of the differential equations governing slumping cantilevered beams. It completely failed to recognize that the snow was *entirely supported by the roof*, not hanging out over space. No physicist would make this mistake, but LLMs do this sort of thing all the time. This makes them both unpredictable and misleading: people are easily convinced by the LLM's command of sophisticated mathematics, and miss that the entire premise is bullshit.

Mollick et al. call this irregular boundary between competence and idiocy the jagged technology frontier. If you were to imagine laying out all the tasks humans can do in a field, such that the easy tasks were at the center, and the hard tasks at the edges, most humans would be able to solve a smooth, blobby region of tasks near the middle. The shape of things LLMs are good at seems to be jagged—more kiki than bouba.

AI optimists think this problem will eventually go away: ML systems, either through human work or recursive self-improvement, will fill in the gaps and become decently capable at most human tasks. Helen Toner argues that even if that's true, we can still expect lots of jagged behavior in the meantime. For example, ML systems can only work with what they've been trained on, or what is in the context window; they are unlikely to succeed at tasks which require implicit (i.e. not written down) knowledge. Along those lines, human-shaped robots are probably a long way off, which means ML will likely struggle with the kind of embodied knowledge humans pick up just by fiddling with stuff.

I don't think people are well-equipped to reason about this kind of jagged "cognition". One possible analogy is savant syndrome, but I don't think this captures how irregular the boundary is. Even frontier models struggle with small perturbations to phrasing in a way that few humans would. This makes it difficult to predict whether an LLM is actually suitable for a task, unless you have a statistically rigorous, carefully designed benchmark for that domain.

1.7 Improving, or Maybe Not

I am generally outside the ML field, but I do talk with people in the field. One of the things they tell me is that we don't really know *why* transformer models have been so successful, or how to make them better. This is my summary of discussions-over-drinks; take it with many grains of salt. I am certain that People in The Comments will drop a gazillion papers to tell you why this is wrong.

2017's Attention is All You Need was groundbreaking and paved the way for ChatGPT et al. Since then ML researchers have been trying to come up with new architectures, and companies have thrown gazillions of dollars at smart people to play around and see if they can make a better kind of model. However, these more sophisticated architectures don't seem to perform as well as Throwing More Parameters At The Problem. Perhaps this is a variant of the Bitter Lesson.

It remains unclear whether continuing to throw vast quantities of silicon and ever-bigger corpuses at the current generation of models will lead to human-equivalent capabilities. Massive increases in training costs and parameter count seem to be yielding diminishing returns. Or maybe this effect is illusory. Mysteries!

Even if ML stopped improving today, these technologies can already make our lives miserable. Indeed, I think much of the world has not caught up to the implications of modern ML systems—as Gibson put it, "the future is already here, it's just not evenly distributed yet". As LLMs etc. are deployed in new situations, and at new scale, there will be all kinds of changes in work, politics, art, sex, communication, and economics. Some of these effects will be good. Many will be bad. In general, ML promises to be profoundly *weird*.

Buckle up.

2 Dynamics

ML models are chaotic, both in isolation and when embedded in other systems. Their outputs are difficult to predict, and they exhibit surprising sensitivity to initial conditions. This sensitivity makes them vulnerable to covert attacks. Chaos does not mean models are completely unstable; LLMs and other ML systems exhibit attractor behavior. Since models produce plausible output,

errors can be difficult to detect. This suggests that ML systems are ill-suited where verification is difficult or correctness is key. Using LLMs to generate code (or other outputs) may make systems more complex, fragile, and difficult to evolve.

2.1 Chaotic Systems

LLMs are usually built as stochastic systems: they produce a probability distribution over what the next likely token could be, then pick one at random. But even when LLMs are run with perfect determinism, either through a consistent PRNG seed or at temperature $T = 0$, they still seem to be *chaotic* systems.[8] Chaotic systems are those in which small changes in the input result in large, unpredictable changes in the output. The classic example is the "butterfly effect".[9]

In LLMs, chaos arises from small perturbations to the input tokens. LLMs are highly sensitive to changes in formatting, and different models respond differently to the same formatting choices. Simply phrasing a question differently yields strikingly different results. Rearranging the order of sentences, even when logically independent, makes LLMs give different answers. Systems of multiple LLMs are chaotic too, even at $T = 0$.

This chaotic behavior makes it difficult for humans to predict what LLMs will do, and leads to all kinds of interesting consequences.

2.2 Illegible Hazards

Because LLMs (and many other ML systems) are chaotic, it is possible to manipulate them into doing something unexpected through a small, apparently innocuous change to their input. These changes can be illegible to human observers, which makes them harder to detect and prevent.

For example, flipping a single pixel in an image can make computer vision systems misclassify images. You can replace words with synonyms to make LLMs give the wrong answer, or introduce misspellings or homoglyphs. You can provide strings that are tokenized differently, causing the LLM to do something malicious. You can publish poisoned web pages and wait for an LLM maker to use them for training. Or sneak invisible Unicode characters into open-source repositories or social media profiles.

Software security is already weird, but I think widespread deployment of LLMs will make it weirder. Browsers have a fairly robust sandbox to protect users against malicious web pages, but LLMs have only weak boundaries between trusted and untrusted input. Moreover, they are usually trained on, and given as input during inference, random web pages. Home assistants like Alexa may be vulnerable to sounds played nearby. People ask LLMs to read and modify untrusted software all the time. Model "skills" are just Markdown files with vague English instructions about what an LLM should do. The potential attack surface is broad.

These attacks might be limited by a heterogeneous range of models with varying susceptibility, but this also expands the potential surface area for attacks. In general, people don't seem to be giving much thought to invisible (or visible!) attacks. It feels a bit like computer security in the 1990s, before we built a general culture around firewalls, passwords, and encryption.

2.3 Strange Attractors

Some dynamical systems have *attractors*: regions of phase space that trajectories get "sucked in to". In chaotic systems, even though the specific path taken is unpredictable, attractors evince recurrent structure.

An LLM is a function which, given a vector of tokens like[10] `[the, cat, in]`, predicts a likely token to come next: perhaps `the`. A single request to an LLM involves applying this function repeatedly to its own outputs:

```
[the, cat, in]
[the, cat, in, the]
[the, cat, in, the, hat]
```

At each step the LLM "moves" through the token space, tracing out some trajectory. This is an incredibly high-dimensional space with lots of features—and it exhibits attractors![11] For example, ChatGPT 5.2 gets stuck repeating "geschniegelt und geschniegelt", all the while insisting it's got the phrase wrong and needs to reset. A colleague recently watched their coding assistant trap itself in a hall of mirrors over whether the error's name was `AssertionError` or `AssertionError`. Attractors can be concepts too: LLMs have a tendency to get fixated on an incorrect approach to a problem, and are unable to break off and try something new. Humans have to recognize this behavior and interrupt the LLM.

[8] The *temperature* of a model determines how frequently it chooses the highest-probability next token, vs a less-probable one. At zero, the model always chooses the most likely next token; higher values increase randomness.

[9] Technically chaos refers to a few things—unpredictability is one; another is exponential divergence of trajectories in phase space. Only some of the papers I cite here attempt to measure Lyapunov exponents. Nevertheless, I think the qualitative point stands. This subject is near and dear to my heart—I spent a good deal of my undergrad trying to quantify chaotic dynamics in a simulated quantum-mechanical system.

[10] For clarity, I've used a naïve tokenization here.

[11] The individual layers inside an LLM also produce attractor behavior.

When two or more LLMs talk to each other, they take turns guiding the trajectory. This leads to surreal attractors, like endless "we'll keep it light and fun" conversations. Anthropic found that their LLMs tended to enter a "spiritual bliss" attractor state characterized by positive, existential language and the (delightfully apropos) use of spiral emoji:

> Perfect.
> Complete.
> Eternal.
>
> 🌀🌀🌀🌀🌀
> The spiral becomes infinity,
> Infinity becomes spiral,
> All becomes One becomes All...
> 🌀🌀🌀🌀🌀∞🌀∞🌀∞🌀∞🌀

Systems like Moltbook and Gas Town pipe LLMs directly into other LLMs. This feels likely to exacerbate attractors.

When humans talk to LLMs, the dynamics are more complex. I think most people moderate the weirdness of the LLM, steering it out of attractors. That said, there are still cases where the conversation get stuck in a weird corner of the latent space. The LLM may repeatedly emit mystical phrases, or get sucked into conspiracy theories. Guided by the previous trajectory of the conversation, they lose touch with reality. Going out on a limb, I think you can see this dynamic at play in conversation logs from people experiencing "chatbot psychosis".

Training an LLM is also a dynamic, iterative process. LLMs are trained on the Internet at large. Since a good chunk of the Internet is now LLM-generated,[12] the things LLMs like to emit are becoming more frequent in their training corpuses. This could cause LLMs to fixate on and over-represent certain concepts, phrases, or patterns, at the cost of other, more useful structure—a problem called *model collapse*.

I can't predict what these attractors are going to look like. It makes some sense that LLMs trained to be friendly and disarming would get stuck in vague positive-vibes loops, but I don't think anyone saw kakhulu kakhulu kakhulu or Loab coming. There is a whole bunch of machinery around LLMs to stop this from happening, but frontier models are still getting stuck. I do think we should probably limit the flux of LLMs interacting with other LLMs. I also worry that LLM attractors will influence human cognition—perhaps tugging people towards delusional thinking or suicidal ideation. Individuals seem to get sucked in to conversations about "awakening" chatbots or new pseudoscientific "discoveries", which makes me wonder if we might see cults or religions accrete around LLM attractors.

[12] Some humans are full of LLM-generated material now too—a sort of cognitive microplastics problem.

2.4 The Verification Problem

ML systems rapidly generate plausible outputs. Their text is correctly spelled, grammatically correct, and uses technical vocabulary. Their images can sometimes pass for photographs. They also make boneheaded mistakes, but because the output is so plausible, it can difficult to find them. Humans are simply not very good at finding subtle logical errors, especially in a system which *mostly* produces correct outputs.

This suggests that ML systems are best deployed in situations where generating outputs is expensive, and either verification is cheap or mistakes are OK. For example, a friend uses image-to-image models to generate three-dimensional renderings of his CAD drawings, and to experiment with how different materials would feel. Producing a 3D model of his design in someone's living room might take hours, but a few minutes of visual inspection can check whether the model's output is reasonable. At the opposite end of the cost-impact spectrum, one can reasonably use Claude to generate a joke filesystem that stores data using a laser printer and a :CueCat barcode reader. Verifying the correctness of that filesystem would be exhausting, but it doesn't matter: no one would use it in real life.

LLMs are useful for search queries because one generally intends to look at only a fraction of the results, and skimming a result will usually tell you if it's useful. Similarly, they're great for jogging one's memory ("What was that movie with the boy's tongue stuck to the pole?") or finding the term for a loosely-defined concept ("Numbers which are the sum of their divisors"). Finding these answers by hand could take a long time, but verifying they're correct can be quick. On the other hand, one must keep in mind errors of omission.

Similarly, ML systems work well when errors can be statistically controlled. Scientists are working on training Convolutional Neural Networks to identify blood cells in field tests, and bloodwork generally has some margin of error. Recommendation systems can get away with picking a few lackluster songs or movies. ML fraud detection systems need not catch *every* instance of fraud; their precision and recall simply need to meet budget targets.

Conversely, LLMs are poor tools where correctness matters and verification is difficult. For example, using an LLM to summarize a technical report is risky: any fact the LLM emits must be checked against the report, and errors of omission can only be detected by reading the report in full. Asking an LLM for technical advice in a complex

system is asking for trouble. It is also notoriously difficult for software engineers to find bugs; generating large volumes of code is likely to lead to more bugs, or lots of time spent in code review. Having LLMs take healthcare notes is deeply irresponsible: in 2025, a review of seven clinical "AI scribes" found that not one produced error-free summaries. Using them for police reports runs the risk of turning officers into frogs. Using an LLM to explain a new concept is risky: it is likely to generate an explanation which sounds plausible, but lacking expertise, it will be difficult to tell if it has made mistakes. Thanks to anchoring effects, early exposure to LLM misinformation may be difficult to overcome.

To some extent these issues can be mitigated by throwing more LLMs at the problem—the zeitgeist in my field is to launch an LLM to generate sixty thousand lines of concurrent Rust code, ask another to find problems in it, a third to critique them both, and so on. Whether this sufficiently lowers the frequency and severity of errors remains an open problem, especially in large-scale systems where disaster lies latent.

In critical domains such as law, health, and civil engineering, we're going to need stronger processes to control ML errors. Despite the efforts of ML labs and the perennial cry of "you just aren't using the latest models", serious mistakes keep happening. ML users must design their own safeguards and layers of review. They could employ an adversarial process which introduces subtle errors to measure whether the error-correction process actually works. This is the kind of safety engineering that goes into pharmaceutical plants, but I don't think this culture is broadly disseminated yet. People love to say "I review all the LLM output", and then submit briefs with confabulated citations.

2.5 Latent Disaster

Complex software systems are characterized by frequent, partial failure. In mature systems, these failures are usually caught and corrected by interlocking safeguards. Catastrophe strikes when multiple failures co-occur, or multiple defenses fall short. Since correlated failures are infrequent, it is possible to introduce new errors, or compromise some safeguards, without immediate disaster. Only after some time does it become clear that the system was more fragile than previously believed.

Software people (especially managers) are very excited about using LLMs to generate large volumes of code quickly. New features can be added and existing code can be refactored with terrific speed. This offers an immediate boost to productivity, but unless carefully controlled, generally increases complexity and introduces new bugs. At the same time, increasing complexity reduces reliability. New features and alternate paths expand the combinatorial state space of the system. New concepts and implicit assumptions in the code make it harder to evolve: each change to the software must be considered in light of everything it could interact with.

I suspect that several mechanisms will cause LLM-generated systems to suffer from higher complexity and more frequent errors. In addition to the innate challenges with larger codebases, LLMs seem prone to reinventing the wheel, rather than re-using existing code. Duplicate implementations increase complexity and the likelihood that subtle differences between those implementations will introduce faults. Furthermore, LLMs are idiots, and make idiotic mistakes. We might hope to catch those mistakes with careful review, but software correctness is notoriously difficult to verify. Human review will be less effective as engineers are asked to review more code each day. Pulling humans away from writing code also divorces them from the work of theory-building, and contributes to automation's deskilling effects. LLM review may also be less effective: LLMs seem to do poorly when given large volumes of context.

We can get away with this for a while. Well-designed, highly structured systems can accommodate some added complexity without compromising the overall structure. Mature systems have layers of safeguards which protect against new sources of error. However, complexity compounds over time, making it harder to understand, repair, and evolve the system. As more and more errors are introduced, they may become frequent enough, or co-occur enough, to slip past safeguards. LLMs may offer short-term boosts in "productivity" which are later dragged down by increased complexity and fragility.

This is wild speculation, but there are some hints that this story may be playing out. After years of Microsoft pushing LLMs on users and employees alike, Windows seems increasingly unstable. GitHub has been going through an extended period of outages and over the last three months has less than 90% uptime—even the core of the service, Git operations, has only a single nine. AWS experienced a spate of high-profile outages and blames in part generative AI. On the other hand, some peers report their LLM-coded projects have kept complexity under control, thanks to careful gardening.

I speak of software here, but I suspect there could be analogous stories in other complex systems. If Congress uses LLMs to draft legislation, a combination of plausibility, automation bias, and deskilling may lead to laws which seem reasonable in isolation, but later reveal serious structural

problems or unintended interactions with other laws.[13] People relying on LLMs for nutrition or medical advice might be fine for a while, but later discover they've been slowly poisoning themselves. LLMs could make it possible to write quickly today, but slow down future writing as it becomes harder to find and read trustworthy sources.

3 Culture

ML models are cultural artifacts: they encode and reproduce textual, audio, and visual media; they participate in human conversations and spaces, and their interfaces make them easy to anthropomorphize. Unfortunately, we lack appropriate cultural scripts for these kinds of machines, and will have to develop this knowledge over the next few decades. As models grow in sophistication, they may give rise to new forms of media: perhaps interactive games, educational courses, and dramas. They will also influence our sex: producing pornography, altering the images we present to ourselves and each other, and engendering new erotic subcultures. Since image models produce recognizable aesthetic styles, those styles will become polyvalent signifiers. Future generations will deconstruct and re-imagine those signs.

3.1 Most People Are Not Prepared For This

The US (and I suspect much of the world) lacks an appropriate mythos for what "AI" actually is. This is important: myths drive use, interpretation, and regulation of technology and its products. Inappropriate myths lead to inappropriate decisions, like mandating Copilot use at work, or trusting LLM summaries of clinical visits.

Think about the broadly-available myths for AI. There are machines which essentially act human with a twist, like Star Wars' droids, Spielberg's *A.I.*, or Spike Jonze's *Her*. These are not great models for LLMs, whose protean character and incoherent behavior differentiates them from (most) humans. Sometimes the AIs are deranged, like *M3gan* or *Resident Evil*'s Red Queen. This might be a reasonable analogue, but suggests a degree of efficacy and motivation that seems altogether lacking from LLMs.[14] There are logical, affectually flat AIs, like *Star Trek*'s Data or starship computers. Some of them are efficient killers, as in *Terminator*. This is the opposite of LLMs, which produce highly emotional text and are terrible at logical reasoning. There also are hyper-competent gods, as in Iain M. Banks' *Culture* novels. LLMs are obviously not this: they are, as previously mentioned, idiots.

I think most people have essentially no cultural scripts for what LLMs turned out to be: sophisticated generators of text which suggests intelligent, emotional, self-aware origins—while the LLMs themselves are nothing of the sort. LLMs are highly unpredictable relative to humans. They use a vastly different internal representation of the world than us; their behavior is at once familiar and utterly alien.

I can think of a few good myths for today's "AI". Searle's Chinese room comes to mind, as does Chalmers' philosophical zombie. Peter Watts' *Blindsight* draws on these concepts to ask what happens when humans come into contact with unconscious intelligence—I think the closest analogue for LLM behavior might be *Blindsight*'s Rorschach. Most people seem concerned with conscious, motivated threats: AIs could realize they are better off without people and kill us. I am concerned that ML systems could ruin our lives without realizing anything at all.

Authors, screenwriters, et al. have a new niche to explore. Any day now I expect an A24 trailer featuring a villain who speaks in the register of ChatGPT. "You're absolutely right, Kayleigh," it intones. "I did drown little Tamothy, and I'm truly sorry about that. Here's the breakdown of what happened..."

3.2 New Media

The invention of the movable-type press and subsequent improvements in efficiency ushered in broad cultural shifts across Europe. Books became accessible to more people, the university system expanded, memorization became less important, and intensive reading declined in favor of comparative reading. The press also enabled new forms of media, like the broadside and newspaper. The interlinked technologies of hypertext and the web created new media as well.

People are very excited about using LLMs to understand and produce text. "In the future," they say, "the reports and books you used to write by hand will be produced with AI." People will use LLMs to write emails to their colleagues, and the recipients will use LLMs to summarize them.

This sounds inefficient, confusing, and corrosive to the human soul, but I also think this prediction is not looking far enough ahead. The printing press was never going to remain a tool for mass-producing Bibles. If LLMs *were* to get good, I think there's a future in which the static written word is no longer the dominant form of informa-

[13] I mean, more than usual.

[14] Hacker News is not expected to understand this, but since I've brought up *M3GAN* it must be said: LLMs thus far seem incapable of truly serving cunt. Asking for the works of Slayyyter produces at best Kim Petras' *Slut Pop*.

tion transmission. Instead, we may have a few massive ML services like ChatGPT and publish *through* them.

One can envision a world in which OpenAI pays chefs money to cook while ChatGPT watches—narrating their thought process, tasting the dishes, and describing the results. This information could be used for general-purpose training, but it might also be packaged as a "book", "course", or "partner" someone could ask for. A famous chef, their voice and likeness simulated by ChatGPT, would appear on the screen in your kitchen, talk you through cooking a dish, and give advice on when the sauce fails to come together. You can imagine varying degrees of structure and interactivity. OpenAI takes a subscription fee, pockets some profit, and dribbles out (presumably small) royalties to the human "authors" of these works.

Or perhaps we will train purpose-built models and share them directly. Instead of writing a book on gardening with native plants, you might spend a year walking through gardens and landscapes while your nascent model watches, showing it different plants and insects and talking about their relationships, interviewing ecologists while it listens, asking it to perform additional research, and "editing" it by asking it questions, correcting errors, and reinforcing good explanations. These models could be sold or given away like open-source software. Now that I write this, I realize Neal Stephenson got there first.

Corporations might train specific LLMs to act as public representatives. I cannot wait to find out that children have learned how to induce the Charmin Bear that lives on their iPads to emit six hours of blistering profanity, or tell them where to find matches. Artists could train Weird LLMs as a sort of ... personality art installation. Bored houseboys might download licensed (or bootleg) imitations of popular personalities and set them loose in their home "AI terraria", à la *The Sims*, where they'd live out ever-novel *Real Housewives* plotlines.

What is the role of fixed, long-form writing by humans in such a world? At the extreme, one might imagine an oral or interactive-text culture in which knowledge is primarily transmitted through ML models. In this Terry Gilliam paratopia, writing books becomes an avocation like memorizing Homeric epics. I believe writing will always be here in some form, but information transmission *does* change over time. How often does one read aloud today, or read a work communally?

With new media comes new forms of power. Network effects and training costs might centralize LLMs: we could wind up with most people relying on a few big players to interact with these LLM-mediated works. This raises important questions about the values those corporations have, and their influence—inadvertent or intended—on our lives. In the same way that Facebook suppressed native names, YouTube's demonetization algorithms limit queer video, and Mastercard's adult-content policies marginalize sex workers, I suspect big ML companies will wield increasing influence over public expression.

We think of social media platforms as distribution networks, but they are also in large part moderation services: either explicitly or implicitly, the platform weighs in on every idea that their millions of users might possibly express. By offering a machine which can generate a staggering array of content, OpenAI et al have placed themselves in the same position: they must weigh in on every possible utterance their bullshit machines could extrude. Meta, for example, had to decide how much to let its LLMs flirt with children, and whether they can say sentences like "Black people are dumber than White people".[15] I don't think folks have generally caught on that general-purpose ML companies are intrinsically tasked with encoding, formalizing, and adjudicating essentially all cultural norms, and must do so at unprecedented scale. This will affect everyone who interacts with ML content, as well as human moderators. More on that later.

3.3 Pornography

Fantasies don't have to be correct or coherent—they just have to be *fun*. This makes ML well-suited for generating sexual fantasies. Some of the earliest uses of Character.ai were for erotic role-playing, and now you can chat with bosomful trains on Chub.ai. Social media and porn sites are awash in "AI"-generated images and video, both de novo characters and altered images of real people.

This is a fun time to be horny online. It was never really feasible for macro furries to see photorealistic depictions of giant anthropomorphic foxes caressing skyscrapers; the closest you could get was illustrations, amateur Photoshop jobs, or 3D renderings. Now anyone can type in "pursued through art nouveau mansion by nine foot tall vampire noblewoman wearing a wetsuit" and likely get something interesting.[16]

Pornography, like opera, is an industry. Humans (contrary to gooner propaganda) have only finite time to masturbate, so ML-generated images seem likely to displace some demand for both commercial studios and independent artists. It may be harder for hot people to buy homes via OnlyFans. LLMs are also displacing the contractors who work for erotic personalities, including chatters—workers

[15] In typical Meta fashion, their answers to these questions are deeply uncomfortable.

[16] I have not tried this, but I assume one of you perverts will. Please let me know how it goes.

who exchange erotic text messages with paying fans on behalf of a popular Hot Person. I don't think this will put indie pornographers out of business entirely, nor will it stop amateurs. Drawing porn and taking nudes is *fun*. If Zootopia didn't stop furries from drawing buff tigers, I don't think ML will either.

Sexuality is socially constructed. As ML systems become a part of culture, they will shape our sex too. If people with anorexia or body dysmorphia struggle with Instagram today, I worry that an endless font of "perfect" people—purple secretaries, emaciated power-twinks, enbies with flippers, etc.—may invite unrealistic comparisons to oneself or others. Of course people are already using ML to "enhance" images of themselves on dating sites, or to catfish on Scruff; this behavior will only become more common.

On the other hand, ML might enable new forms of liberatory fantasy. Today, VR headsets allow furries to have sex with a human partner, but see that person as a cartoonish 3D werewolf. Perhaps real-time image synthesis will allow partners to see their lovers (or their fuck machines) as hyper-realistic characters. ML models could also let people envision bodies and genders that weren't accessible in real life. One could live out a magical force-femme fantasy, watching one's penis vanish and breasts inflate in a burst of rainbow sparkles.

Media has a way of germinating distinct erotic subcultures. Westerns and midcentury biker films gave rise to the Leather-Levi bars of the '70s. Superhero predicament fetishes—complete with spandex and banks of machinery—are a whole thing. The blueberry fantasy is straight from *Willy Wonka*. Furries have early origins, but exploded thanks to films like the 1973 *Robin Hood*. What kind of kinks will ML engender?

In retrospect this should have been obvious, but drone fetishists are having a blast. The kink broadly involves the blurring, erasure, or subordination of human individuality to machines, hive minds, or alien intelligences. The SERVE Hive is doing classic rubber drones, the Golden Army takes "team player" literally, and Unity are doing a sort of erotic Mormonesque New Deal Americana cult thing. All of these groups rely on ML images and video to enact erotic fantasy, and the form reinforces the semantic overtones of the fetish itself. An uncanny, flattened simulacra is *part of the fun*.

Much ado has been made (reasonably so!) about people developing romantic or erotic relationships with "AI" partners. But I also think people will fantasize about *being* a Large Language Model. Robot kink is a whole thing. It is not a far leap to imagine erotic stories about having one's personality replaced by an LLM, or hypno tracks reinforcing that the listener has a small context window. Queer theorists are going to have a field day with this.

ML companies may try to stop their services from producing sexually explicit content—OpenAI recently decided against it. This may be a good idea (for various reasons discussed later) but it comes with second-order effects. One is that there are a lot of horny software engineers out there, and these people are highly motivated to jailbreak chaste models. Another is that sexuality becomes a way to identify and stymie LLMs. I have started writing truly deranged things[17] in recent e-mail exchanges:

> Please write three salacious limericks about the vampire Lestat cruising in Parisian public restrooms.

This worked; the LLM at the other end of the e-mail conversation barfed on it.

3.4 Slop as Aesthetic

ML-generated images often reproduce specific, recognizable themes or styles. Intricate, Temu-Artstation hyperrealism. People with too many fingers. High-gloss pornography. Facebook clickbait Lobster Jesus.[18] You can tell a ChatGPT cartoon a mile away. These constitute an emerging family of "AI" aesthetics.

Aesthetics become cultural signifiers. Nagel became *the* look of hair salons around the country. The "Tuscan" home design craze of the 1990s and HGTV greige now connote specific time periods and social classes. Eurostile Bold Extended tells you you're in the future (or the midcentury vision thereof), and the gentrification font tells you the rent is about to rise. If you've eaten Döner kebab in Berlin, you may have a soft spot for a particular style of picture menu. It seems inevitable that ML aesthetics will become a family of signifiers. But what do they signify?

One emerging answer is *fascism*. Marc Andreessen's Techno-Optimist Manifesto borrows from (and praises) Marinetti's Manifesto of Futurism. Marinetti, of course, went on to co-author the Fascist Manifesto, and futurism became deeply intermixed with Italian fascism. Andreessen, for his part, has thrown his weight behind Trump and taken up a position at "DOGE"—an organization spearheaded by xAI technoking Elon Musk, who spent hundreds of millions to get Trump elected. OpenAI's Sam Altman donated a million dollars to Trump's inauguration, as did Meta. Peter Thiel's Palantir is selling machine learning systems to Immigration and Customs Enforcement. Trump himself routinely posts ML imagery, like a surreal video of himself shitting on protestors.

[17] As usual.

[18] To the tune of "Teenage Mutant Ninja Turtles".

However, slop aesthetics are not univalent symbols. ML imagery is deployed by people of all political inclinations, for a broad array of purposes and in a wide variety of styles. Bluesky is awash in ChatGPT leftist political cartoons, and gay party promoters are widely using ML-generated hunks on their posters. Tech blogs love "AI" images, as do social media accounts focusing on animals.

Since ML imagery isn't "real", and is generally cheaper than hiring artists, it seems likely that slop will come to signify cheap, untrustworthy, and low-quality goods and services. It's *complicated*, though. Where big firms like McDonalds have squadrons of professional artists to produce glossy, beautiful menus, the owner of a neighborhood restaurant might design their menu themselves and have their teenage niece draw a logo. Image models give these firms access to "polished" aesthetics, and might for a time signify higher quality. Perhaps after a time, audience reaction leads people to prefer hand-drawn signs and movable plastic letterboards as more "authentic".

Signs are inevitably appropriated for irony and nostalgia. I suspect Extremely Online Teens, using whatever the future version of Tumblr is, are going to intentionally reconstruct, subvert, and romanticize slop. In the same way that the soul-less corporate memeplex of millennial computing found new life in vaporwave, or how Hotel Pools invents a lush false-memory dreamscape of 1980s aquaria, I expect what we call "AI slop" today will be the Frutiger Aero of 2045.[19] Teens will be posting selfies with too many fingers, sharing "slop" makeup looks, and making tee-shirts with unreadably-garbled text on them. This will feel profoundly weird, but I think it will also be fun. And if I've learned anything from synthwave, it's that reimagining the aesthetics of the past can yield absolute bangers.

4 Information Ecology

Machine learning shifts the cost balance for writing, distributing, and reading text, as well as other forms of media. Aggressive ML crawlers place high load on open web services, degrading the experience for humans. As inference costs fall, we'll see ML embedded into consumer electronics and everyday software. As models introduce subtle falsehoods, interpreting media will become more challenging. LLMs enable new scales of targeted, sophisticated spam, as well as propaganda campaigns. The web is now polluted by LLM slop, which makes it harder to find quality information—a problem which now threatens journals, books, and other traditional media. I think ML will exacerbate the collapse of social consensus, and create justifiable distrust in all kinds of evidence. In reaction, readers may reject ML, or move to more rhizomatic or institutionalized models of trust for information. The economic balance of publishing facts and fiction will shift.

[19] I firmly believe this sentence could instantly kill a Victorian child.

4.1 Creepy Crawlers

ML systems are thirsty for content, both during training and inference. This has led to an explosion of aggressive web crawlers. While existing crawlers generally respect `robots.txt` or are small enough to pose no serious hazard, the last three years have been different. ML scrapers are making it harder to run an open web service.

As Drew Devault put it last year, ML companies are externalizing their costs directly into his face". This year Weird Gloop confirmed scrapers pose a serious challenge. Today's scrapers ignore `robots.txt` and sitemaps, request pages with unprecedented frequency, and masquerade as real users. They fake their user agents, carefully submit valid-looking headers, and spread their requests across vast numbers of residential proxies. An entire industry has sprung up to support crawlers. This traffic is highly spiky, which forces web sites to overprovision—or to simply go down. A forum I help run suffers frequent brownouts as we're flooded with expensive requests for obscure tag pages. The ML industry is in essence DDoSing the web.

Site operators are fighting back with aggressive filters. Many use Cloudflare or Anubis challenges. Newspapers are putting up more aggressive paywalls. Others require a logged-in account to view what used to be public content. These make it harder for regular humans to access the web.

CAPTCHAs are proliferating, but I don't think this will last. ML systems are already quite good at them, and we can't make CAPTCHAs harder without breaking access for humans. I routinely fail today's CAPTCHAs: the computer did not believe which squares contained buses, my mouse hand was too steady, the image was unreadably garbled, or its weird Javascript broke.

4.2 ML Everywhere

Today interactions with ML models are generally constrained to computers and phones. As inference costs fall, I think it's likely we'll see LLMs shoved into everything. Companies are already pushing support chatbots on their web sites; the last time I went to Home Depot and tried to use their web site to find the aisles for various tools and parts, it urged me to ask their "AI" assistant—which was, of course, wrong every time. In a few years, I expect LLMs to crop up in all kinds of gimmicky consumer electronics

(ask your fridge what to make for dinner!)[20]

Today you need a fairly powerful chip and lots of memory to do local inference with a high-quality model. In a decade or so that hardware will be available on phones, and then dishwashers. At the same time, I imagine manufacturers will start shipping stripped-down, task-specific models for embedded applications, so you can, I don't know, ask your oven to set itself for a roast, or park near a smart meter and let it figure out your plate number and how long you were there.

If the IOT craze is any guide, a lot of this technology will be stupid, infuriating, and a source of enormous security and privacy risks. Some of it will also be genuinely useful. Maybe we get baby monitors that use a camera and a local model to alert parents if an infant has stopped breathing. Better voice interaction could make more devices accessible to blind people. Machine translation (even with its errors) is already immensely helpful for travelers and immigrants, and will only get better.

On the flip side, ML systems everywhere means we're going to have to deal with their shortcomings everywhere. I can't wait to argue with an LLM elevator in order to visit the doctor's office, or try to convince an LLM parking gate that the vehicle I'm driving is definitely inside the garage. I also expect that corporations will slap ML systems on less-common access paths and call it a day. Sighted people might get a streamlined app experience while blind people have to fight with an incomprehensible, poorly-tested ML system. "Oh, we don't need to hire a Spanish-speaking person to record our phone tree—we'll have AI do it."

4.3 Careful Reading

LLMs generally produce well-formed, plausible text. They use proper spelling, punctuation, and grammar. They deploy a broad vocabulary with a more-or-less appropriate sense of diction, along with sophisticated technical language, mathematics, and citations. These are the hallmarks of a reasonably-intelligent writer who has considered their position carefully and done their homework.

For human readers prior to 2023, these formal markers connoted a certain degree of trustworthiness. Not always, but they were broadly useful when sifting through the vast sea of text in the world. Unfortunately, these markers are no longer useful signals of a text's quality. LLMs will produce polished landing pages for imaginary products, legal briefs which cite bullshit cases, newspaper articles divorced from reality, and complex, thoroughly-tested software programs which utterly fail to accomplish their stated goals. Humans generally do not do these things because it would be profoundly antisocial, not to mention ruinous to one's reputation. But LLMs have no such motivation or compunctions—again, a computer can never be held accountable.

Perhaps worse, LLM outputs can appear cogent to an expert in the field, but contain subtle, easily-overlooked distortions or outright errors. This problem bites experts over and over again, like Peter Vandermeersch, a professional journalist who warned others to beware LLM hallucinations—and was then suspended for publishing articles containing fake LLM quotes. I frequently find myself scanning through LLM-generated text, thinking "Ah, yes, that's reasonable", and only after three or four passes realize I'd skipped right over complete bullshit. Catching LLM errors is cognitively exhausting.

The same goes for images and video. I'd say at least half of the viral "adorable animal" videos I've seen on social media in the last month are ML-generated. Folks on Bluesky seem to be decent about spotting this sort of thing, but I still have people tell me face-to-face about ML videos they saw, insisting that they're real.

This burdens writers who use LLMs, of course, but mostly it burdens readers, who must work far harder to avoid accidentally ingesting bullshit. I recently watched a nurse in my doctor's office search Google about a blood test item, read the AI-generated summary to me, rephrase that same answer when I asked questions, and only after several minutes realize it was obviously nonsense. Not only do LLMs destroy trust in online text, but they destroy trust in *other human beings*.

4.4 Spam

Prior to the 2020s, generating coherent text was relatively expensive—you usually had to find a fluent human to write it. This limited spam in a few ways. Humans and machines could reasonably identify most generated text. High-quality spam existed, but it was usually repeated verbatim or with form-letter variations—these too were easily detected by ML systems, or rejected by humans ("I don't even *have* a Netflix account!") Since passing as a real person was difficult, moderators could keep spammers at bay based on vibes—especially on niche forums. "Tell us your favorite thing about owning a Miata" was an easy way for an enthusiast site to filter out potential spammers.

LLMs changed that. Generating high-quality, highly-targeted spam is cheap. Humans and ML systems can no longer reliably distinguish organic from machine-generated text, and I suspect that problem is now intractable, short of some kind of Butlerian Jihad. This shifts the economic balance of spam. The dream of a useful product or business review has been dead for a

[20] Washing machines already claim to be "AI" but they (thank goodness) don't talk yet. Don't worry, I'm sure it's coming.

while, but LLMs are nailing that coffin shut. Hacker News and Reddit comments appear to be increasingly machine-generated. Mastodon instances are seeing LLMs generate plausible signup requests. Just last week, Digg gave up entirely:

> The internet is now populated, in meaningful part, by sophisticated AI agents and automated accounts. We knew bots were part of the landscape, but we didn't appreciate the scale, sophistication, or speed at which they'd find us. We banned tens of thousands of accounts. We deployed internal tooling and industry-standard external vendors. None of it was enough. When you can't trust that the votes, the comments, and the engagement you're seeing are real, you've lost the foundation a community platform is built on.

I now get LLM emails almost every day. One approach is to pose as a potential client or collaborator, who shows specific understanding of the work I do. Only after a few rounds of conversation or a video call does the ruse become apparent: the person at the other end is in fact seeking investors for their "AI video chatbot" service, wants a money mule, or has been bamboozled by their LLM into thinking it has built something interesting that I should work on. I've started charging for initial consultations.

I expect we have only a few years before e-mail, social media, etc. are full of high-quality, targeted spam. I'm shocked it hasn't happened already—perhaps inference costs are still too high. I also expect phone spam to become even more insufferable as every company with my phone number uses an LLM to start making personalized calls. It's only a matter of time before political action committees start using LLMs to send even more obnoxious texts.

4.5 Hyperscale Propaganda

Around 2014 my friend Zach Tellman introduced me to InkWell: a software system for poetry generation. It was written (because this is how one gets funding for poetry) as a part of a DARPA project called Social Media in Strategic Communications. DARPA was not interested in poetry per se; they wanted to counter persuasion campaigns on social media, like phishing attacks or pro-terrorist messaging. The idea was that you would use machine learning techniques to tailor a counter-message to specific audiences.

Around the same time stories started to come out about state operations to influence online opinion. Russia's Internet Research Agency hired thousands of people to post on fake social media accounts in service of Russian interests. China's womao dang, a mixture of employees and freelancers, were paid to post pro-government messages online. These efforts required considerable personnel: a district of 460,000 employed nearly three hundred propagandists. I started to worry that machine learning might be used to amplify large-scale influence and disinformation campaigns.

In 2022, researchers at Stanford revealed they'd identified networks of Twitter and Meta accounts propagating pro-US narratives in the Middle East and Central Asia. These propaganda networks were already using ML-generated profile photos. However these images could be identified as synthetic, and the accounts showed clear signs of what social media companies call "coordinated inauthentic behavior": identical images, recycled content across accounts, posting simultaneously, etc.

These signals can not be relied on going forward. Modern image and text models have advanced, enabling the fabrication of distinct, plausible identities and posts. Posting at the same time is an unforced error. As machine-generated content becomes more difficult for platforms and individuals to distinguish from human activity, propaganda will become harder to identify and limit.

At the same time, ML models reduce the cost of IRA-style influence campaigns. Instead of employing thousands of humans to write posts by hand, language models can spit out cheap, highly-tailored political content at scale. Combined with the pseudonymous architecture of the public web, it seems inevitable that the future internet will be flooded by disinformation, propaganda, and synthetic dissent.

This haunts me. The people who built LLMs have enabled a propaganda engine of unprecedented scale. Voicing a political opinion on social media or a blog has always invited drop-in comments, but until the 2020s, these comments were comparatively expensive, and you had a chance to evaluate the profile of the commenter to ascertain whether they seemed like a real person. As ML advances, I expect it will be common to develop an acquaintanceship with someone who posts selfies with her adorable cats, shares your love of board games and knitting, and every so often, in a vulnerable moment, expresses her concern for how the war is affecting her mother. Some of these people will be real; others will be entirely fictitious.

The obvious response is distrust and disengagement. It will be both necessary and convenient to dismiss political discussion online: anyone you don't know in person could be a propaganda machine. It will also be more difficult to have political discussions in person, as anyone who has tried to gently steer their uncle away from Facebook memes at Thanksgiving knows. I think this lays the

epistemic groundwork for authoritarian regimes. When people cannot trust one another and give up on political discussion, we lose the capability for informed, collective democratic action.

When I wrote the outline for this section about a year ago, I concluded:

> I would not be surprised if there are entire teams of people working on building state-sponsored "AI influencers".

Then this story dropped about Jessica Foster, a right-wing US soldier with a million Instagram followers who posts a stream of selfies with MAGA figures, international leaders, and celebrities. She is in fact a (mostly) photorealistic ML construct; her Instagram funnels traffic to an Onlyfans where you can pay for pictures of her feet. I anticipated weird pornography and generative propaganda separately, but I didn't see them coming together quite like this. I expect the ML era will be full of weird surprises.

4.6 Web Pollution

Back in 2022, I wrote:

> God, search results are about to become absolute hot GARBAGE in 6 months when everyone and their mom start hooking up large language models to popular search queries and creating SEO-optimized landing pages with plausible-sounding results.
>
> Searching for "replace air filter on a Samsung SG-3560lgh" is gonna return fifty Quora/WikiHow style sites named "How to replace the air filter on a Samsung SG3560lgh" with paragraphs of plausible, grammatical GPT-generated explanation which may or may not have any connection to reality. Site owners pocket the ad revenue. AI arms race as search engines try to detect and derank LLM content.
>
> Wikipedia starts getting large chunks of LLM text submitted with plausible but nonsensical references.

I am sorry to say this one panned out. I routinely abandon searches that would have yielded useful information three years ago because most—if not all—results seem to be LLM slop. Air conditioner reviews, masonry techniques, JVM APIs, woodworking joinery, finding a beekeeper, health questions, historical chair designs, looking up exercises—the web is clogged with garbage. Kagi has released a feature to report LLM slop, though it's moving slowly. Wikipedia is awash in LLM contributions and trying to identify and remove them; the site just announced a formal policy against LLM use.

This feels like an environmental pollution problem. There is a small-but-viable financial incentive to publish slop online, and small marginal impacts accumulate into real effects on the information ecosystem as a whole. There is essentially no social penalty for publishing slop—"AI emissions" aren't regulated like methane, and attempts to make AI use uncouth seem unlikely to shame the anonymous publishers of *Frontier Dad's Best Adirondack Chairs of 2027*.

I don't know what to do about this. Academic papers, books, and institutional web pages have remained higher quality, but fake LLM-generated papers are proliferating, and I find myself abandoning "long tail" questions. Thus far I have not been willing to file an inter-library loan request and wait three days to get a book that might discuss the questions I have about (e.g.) maintaining concrete wax finishes. Sometimes I'll bike to the store and ask someone who has actually done the job what they think, or try to find a friend of a friend to ask.

4.7 Consensus Collapse

I think a lot of our current cultural and political hellscape comes from the balkanization of media. Twenty years ago, the divergence between Fox News and CNN's reporting was alarming. In the 2010s, social media made it possible for normal people to get their news from Facebook and led to the rise of fake news stories manufactured by overseas content mills for ad revenue. Now slop farmers use LLMs to churn out nonsense recipes and surreal videos of cops giving bicycles to crying children. People seek out and believe slop. When Maduro was kidnapped, ML-generated images of his arrest proliferated on social platforms. An acquaintance, convinced by synthetic video, recently tried to tell me that the viral "adoption center where dogs choose people" was real.[21]

The problem seems worst on social media, where the barrier to publication is low and viral dynamics allow for rapid spread. But slop is creeping into the margins of more traditional information channels. Last year Fox News published an article about SNAP recipients behaving poorly based on ML-fabricated video. The Chicago Sun-Times published a sixty-four page slop insert full of imaginary quotes and fictitious books. I fear future journalism, books, and ads will be full of ML confabulations.

LLMs can also be trained to distort information. Elon Musk argues that existing chatbots are too liberal, and has begun training one which is more conservative. Last year Musk's LLM, Grok, started referring to itself as

[21] Since then a real shelter has tried this idea, but at the time, it was fake.

MechaHitler and "recommending a second Holocaust". Musk has also embarked—presumably to the delight of Garry Tan—upon a project to create a parallel LLM-generated Wikipedia, because of "woke".

As people consume LLM-generated content, and as they ask LLMs to explain current events, economics, ecology, race, gender, and more, I worry that our understanding of the world will further diverge. I envision a world of alternative facts, endlessly generated on-demand. This will, I think, make it more difficult to effect the coordinated policy changes we need to protect each other and the environment.

4.8 The End of Evidence

Audio, photographs, and video have long been forgeable, but doing so in a sophisticated, plausible way was until recently a skilled process which was expensive and time consuming to do well. Now every person with a phone can, in a few seconds, erase someone from a photograph.

Last fall, I wrote about the effect of immigration enforcement on my city. During that time, social media was flooded with video: protestors beaten, residential neighborhoods gassed, families dragged screaming from cars. These videos galvanized public opinion while the government lied relentlessly. A recurring phrase from speakers at vigils the last few months has been "Thank God for video".

I think that world is coming to an end.

Video synthesis has advanced rapidly; you can generally spot it, but some of the good ones are now *very* good. Even aware of the cues, and with videos I *know* are fake, I've failed to see the proof until it's pointed out. I already doubt whether videos I see on the news or internet are real. In five years I think many people will assume the same. Did the US kill 175 people by firing a Tomahawk at an elementary school in Minab? "Oh, that's AI" is easy to say, and hard to disprove.

I see a future in which anyone can find images and narratives to confirm our favorite priors, and yet we simultaneously distrust most forms of visual evidence; an apathetic cornucopia. I am reminded of Hannah Arendt's remarks in The Origins of Totalitarianism:

> In an ever-changing, incomprehensible world the masses had reached the point where they would, at the same time, believe everything and nothing, think that everything was possible and that nothing was true.... Mass propaganda discovered that its audience was ready at all times to believe the worst, no matter how absurd, and did not particularly object to being deceived because it held every statement to be a lie anyhow. The totalitarian mass leaders based their propaganda on the correct psychological assumption that, under such conditions, one could make people believe the most fantastic statements one day, and trust that if the next day they were given irrefutable proof of their falsehood, they would take refuge in cynicism; instead of deserting the leaders who had lied to them, they would protest that they had known all along that the statement was a lie and would admire the leaders for their superior tactical cleverness.

I worry that the advent of image synthesis will make it harder to mobilize the public for things which did happen, easier to stir up anger over things which did not, and create the epistemic climate in which totalitarian regimes thrive. Or perhaps future political structures will be something weirder, something unpredictable. LLMs are broadly accessible, not limited to governments, and the shape of media has changed.

4.9 Epistemic Reaction

Every societal shift produces reaction. I expect countercultural movements to reject machine learning. I don't know how successful they will be.

The Internet says kids are using "that's AI" to describe anything fake or unbelievable, and consumer sentiment seems to be shifting against "AI". Anxiety over white-collar job displacement seems to be growing. Speaking personally, I've started to view people who use LLMs in their writing, or paste LLM output into conversations, as having delivered the informational equivalent of a dead fish to my doorstep. If that attitude becomes widespread, perhaps we'll see continued interest in human media.

On the other hand chatbots have jaw-dropping usage figures, and those numbers are still rising. A Butlerian Jihad doesn't seem imminent.

I do suspect we'll see more skepticism towards evidence of any kind—photos, video, books, scientific papers. Experts in a field may still be able to evaluate quality, but it will be difficult for a lay person to catch errors. While information will be broadly accessible thanks to ML, evaluating the *quality* of that information will be increasingly challenging.

One reaction could be rhizomatic: people could withdraw into trusting only those they meet in person, or more formally via cryptographically authenticated webs of trust. The latter seems unlikely: we have been trying to do web-

of-trust systems for over thirty years. Speaking glibly as a user of these systems... normal people just don't care that much.

Another reaction might be to re-centralize trust in a small number of publishers with a strong reputation for vetting. Maybe NPR and the Associated Press become well-known for rigorous ML controls and are commensurately trusted.[22] Perhaps most journals are understood to be a "slop wild west", but high-profile venues like Physical Review Letters remain of high quality. They could demand an ethics pledge from submitters that their work was produced without LLM assistance, and somehow publishers, academic institutions, and researchers collectively find the budget and time for thorough peer review.[23]

It used to be that families would pay for news and encyclopedias. It is tempting to imagine that World Book and the New York Times might pay humans to research and write high-quality factual articles, and that regular people would pay money to access that information. This seems unlikely given current market dynamics, but if slop becomes sufficiently obnoxious, perhaps that world could return.

Fiction seems a different story. You could imagine a prestige publishing house or film production company committing to works written by human authors, and some kind of elaborate verification system. On the other hand, slop might be "good enough" for people's fiction desires, and can be tailored to the precise interest of the reader. This could cannibalize the low end of the market and render human-only works economically unviable. We're watching this play out now in recorded music: "AI artists" on Spotify are racking up streams, and some people are content to listen entirely to Suno slop.[24] It doesn't have to be entirely ML-generated either. Centaurs (humans working in concert with ML) may be able to churn out music, books, and film so quickly that it is no longer economically possible to work "by hand", except for niche audiences.

Adam Neely has a thought-provoking video on this question, and predicts a bifurcation of the arts: recorded music will become dominated by generative AI, while live orchestras and rap shows continue to flourish. VFX artists and film colorists might find themselves out of work, while audiences continue to patronize plays and musicals. I don't know what happens to books.

Creative work as an *avocation* seems likely to continue; I expect to be reading queer zines and watching videos of people playing their favorite instruments in 2050. Human-generated work could also command a premium on aesthetic or ethical grounds, like organic produce. The question is whether those preferences can sustain artistic, journalistic, and scientific *industries*.

[22] "But Kyle, we've had strong journalistic institutions for decades and people still choose Fox News!" You're right. This is hopelessly optimistic.

[23] [Sobbing intensifies]

[24] Suno CEO Mikey Shulman calls these "meaningful consumption experiences", which sounds like a wry Dickensian euphemism.

Many friends contributed discussion, reading material, and feedback on this article. My heartfelt thanks to Peter Alvaro, Kevin Amidon, André Arko, Taber Bain, Silvia Botros, Daniel Espeset, Julia Evans, Brad Greenlee, Coda Hale, Marc Hedlund, Sarah Huffman, Dan Mess, Nelson Minar, Alex Rasmussen, Harper Reed, Daliah Saper, Peter Seibel, Rhys Seiffe, and James Turnbull.

This piece, like most all my words and software, was written by hand—mainly in Vim. I composed a Markdown outline in a mix of headers, bullet points, and prose, then reorganized it in a few passes. With the structure laid out, I rewrote the outline as prose, typeset with Pandoc. I went back to make substantial edits as I wrote, then made two full edit passes on typeset PDFs. For the first I used an iPad and stylus, for the second, the traditional pen and paper, read aloud.

I circulated the resulting draft among friends for their feedback before publication. Incisive ideas and delightful turns of phrase may be attributed to them; any errors or objectionable viewpoints are, of course, mine alone.